



LAKIREDDY BALI REDDY COLLEGE OF ENGINEERING, (AUTONOMOUS)

Approved by AICTE, Affiliated to JNTUK, Accredited by NBA,
ISO 9001:2008 Certified & Accredited by NAAC with "A" Grade

EXAMINATION SECTION

M.Tech. I Semester Supplementary Semester End Examinations, NOVEMBER/DECEMBER, 2013

TIME : 10.00 AM TO 01.00 PM

DATE	(CSE) Computer Science and Engineering	(IT) Software Engineering	(ECE) Systems and Signal Processing	(ME) Thermal Engineering	(EEE) Power Electronics and Drives
26-11-2013 (Tuesday)	MCS101 Data Structures & Algorithm Analysis	MIT101 Requirements Engg. & Estimation	MEC101 Advanced Digital Signal Processing	MME101 Advanced Mathematics	MEE101 Machine Modeling and Analysis
28-11-2013 (Thursday)	MCS102 Database Management Systems	MIT102 Software Architectures	MEC102 Transform Techniques	MME102 Advanced Thermodynamics	MEE102 Power Converters
30-11-2013 (Saturday)	MCS103 Computer Organization	MIT103 Software Process Management	MEC103 VLSI Technology and Design	MME103 Advanced Heat & Mass Transfer	MEE103 Control of Motor Drives-I
03-12-2013 (Tuesday)	MCS104 Computer Networks	MIT104 Web Searching and Mining	MEC104 Microcontrollers For Embedded System Design	MME104 IC Engine Combustion & Pollution	MEE104 Special machines
05-12-2013 (Thursday)	<u>Elective-I</u> MCS1051 Software Engineering	<u>Elective-I</u> MIT1052 Mobile Computing	<u>Elective-I</u> MEC1052 Image and Video Processing	<u>Elective-I</u> MME1051 Gas Dynamics	<u>Elective-I</u> MEE1051 Modern Control Theory
07-12-2013 (Saturday)	<u>Elective-II</u> MCS1063 Artificial Intelligence MCS1061 Automata and Compiler Design	<u>Elective-II</u> MIT1063 Digital Image Processing	<u>Elective-II</u> MEC1061 Radar Signal Processing	<u>Elective-II</u> MME1061 Solar Energy	<u>Elective-II</u> MEE1061 Digital controllers for Power Electronic systems

NOTE:

- Any omissions or clashes in this time table may please be informed to the Controller of Examinations immediately.
- Even if government declares holiday on any of the above dates, the examinations shall be conducted as usual.
- For any clarification in respect of the above examinations please contact Controller of Examinations

Date : 06-11-2013

Copy to: 1. M.TECH. Notice Boards 2. M.TECH. H.O.Ds for n.a.,

[Signature]
DIRECTOR

LAKIREDDY BALI REDDY COLLEGE OF ENGINEERING (AUTONOMOUS)
L.B. Reddy Nagar :: Mylavaram – 521 230 :: Krishna Dist.

M.Tech. I Semester Regular/Supplementary Examinations

ADVANCED MATHEMATICS
(Thermal Engineering (ME))

26 NOV 2013

Time : 3 hours

Max. Marks: 60

Answer all the questions
All Questions carry equal marks

Note: For all numerical computation, three decimal accuracy is required
scientific and non programmable calculators are permitted.

- 1(a) Explain the Gauss-Jordan elimination method of solving system of linear equations. Using this method, solve the system of equations.

$$\begin{aligned}x + y + z &= 1 \\4x + 3y - z &= 6 \\3x + 5y + 3z &= 4\end{aligned}$$

[4+8M]

(OR)

- (b) Solve the following system of equations using LU – decomposition method:

$$\begin{aligned}20x + y - 2z &= 17 \\3x + 20y - z &= -18 \\2x - 3y + 20z &= 25\end{aligned}$$

[12M]

- 2(a) Find the largest eigen value and the corresponding eigen vector of the following matrix.

$$A = \begin{bmatrix} 1 & 3 & -1 \\ 3 & 2 & 4 \\ -1 & 4 & 10 \end{bmatrix}$$

[12M]

(OR)

- (b) Find $\frac{dy}{dx}$ at $x = 16$ using Newton's forward difference interpolation formula from the following data:

x: 15	17	19	21	23	25
y: 3.873	4.123	4.359	4.583	4.796	5.8

[12M]

- 3(a) Explain the Richardson's extrapolation method and mention its advantage in numerical methods

$$\text{Compute } I = \int_0^1 \frac{x}{x^3 + 10} dx$$

Using Simpson's rule with 3, 5 and 9 points improve the results using Romberg integration method. [8+4M]

(OR)

- (b) Given that $\frac{dy}{dx} = x^2 + y$; $y(0) = 1$, determine the value of y when $x = 0.1$ using modified Euler's method, by taking $h = 0.05$ [12M]

- 4(a) Solve the boundary value problem

$$y'' - 64y + 10 = 0 ; y(0) = y(1) = 0$$

by finite difference method. Compute the value of $y(0.5)$ with $n = 2;4$ compare your solution with the exact value. [5+5+2M]

(OR)

- (b) Solve the boundary value problem

$u'' = u + x$, $x \in [0,1]$, $u(0) = 0$, $u(1) = 0$, using the shooting method. Use the Runge-Kutta method of second order to solve the initial value problem with $h = 0.2$. [12M]

- 5(a) State the Dirichlets problems

Derive the five point difference scheme for the Dirichlets problem. Write down the five point difference scheme for the poisson equation for a square grid. [2+8+2M]

(OR)

- (b) Solve $\nabla^2 u = 0$ in R , subject to the $u(x,y) = x - y$ on ∂R where R is the region inside the triangle with vertices $(0,0)$, $(7,0)$, $(0,7)$ and ∂R is its boundary, assume that the step length is 2.0 [12M]

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Code: MEC101

LAKIREDDY BALI REDDY COLLEGE OF ENGINEERING (AUTONOMOUS)
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M.Tech. I Semester Supplementary Examinations, October 2012

ADVANCED DIGITAL SIGNAL PROCESSING

(Signals and System Processing)

26 NOV 2013

Time: 3 hours

Max. Marks: 60

Answer all the questions
All Questions carry equal marks

- 1(a) State and prove the circular time shifting property of DFT.
(b) Compute the DFT of the following sequence using DIT-FFT algorithm
 $x(n) = \{1, 1, 1, 1, 1, 1, 1, 1\}$ [6+6M]
(OR)
(c) Explain the fractional sampling rate conversion with time domain and frequency domain relations.
(d) Compare IIR & FIR filters. [6+6M]
- 2(a) Explain the performance measures for the periodogram & Bartlett of Nonparametric methods of power spectrum estimates. [12M]
(OR)
(b) If the sample sequence of a random process has $N = 1000$ samples.
(i) Determine the frequency resolution of Bartlett, Welch (50% overlap) and Blackman – Tukey methods for a quality factor $Q = 10$.
(ii) Determine the record length for the Bartlett, Welch (50% overlap) & Blackman-Tukey methods. [12M]
- 3(a) Explain the power spectrum estimation using Yule-walker method.
(b) Explain ARMA model power spectrum estimation. [6+6M]
(OR)
(c) Determine the relation between the autocorrelation and the model parameters.
(d) List the properties of auto correlation. [6+6M]
- 4(a) Explain forward & Backward linear prediction methods. [12M]
(OR)
(b) An AR(3) process $\{x(n)\}$ is characterised by the auto correlation sequence $r_{xx}(0) = 1$, $r_{xx}(1) = \frac{1}{2}$, $r_{xx}(2) = \frac{1}{8}$ & $r_{xx}(3) = \frac{1}{64}$ using schur algorithm determine the three reflection coefficients k_1 , k_2 & k_3 . [12M]
- 5(a) Explain the effects of quantization in Analog-to-Digital conversion of noise.
(b) Explain the finite word length effects in IIR digital filters. [6+6M]
(OR)
(c) Explain the finite word length effects in fixed point DSP systems. [12M]

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M.Tech. I Semester Regular/Supplementary Examinations

VLSI TECHNOLOGY AND DESIGN
(Systems and Signal Processing (ECE))

30 NOV 2013

Time : 3 hours

Max. Marks: 60

Answer all the questions
All Questions carry equal marks

- 1(a) Draw and explain the n-well process [6M]
(b) Explain the twin tub process with a neat diagram. [6M]
(OR)
(c) Explain the n MOS inverter with the help of transfer characteristics. [6M]
(d) Discuss the small signal model of an MOS transistor. [6M]
- 2(a) Explain CMOS inverter layout plan along with its cross section diagram. [6M]
(b) What are the different scaling models. [6M]
(OR)
(c) Explain the power consumption analysis of static complementary gates with neat diagram. [6M]
(d) Explain about resistive and inductive interconnect delay in gates. [6M]
- 3(a) Explain the testing procedure for faults in combinational networks. [6M]
(b) Draw a layout for a half adder circuit. [6M]
(OR)
(c) Discuss about power optimization techniques. [6M]
(d) Design a layout diagram for a CMOS three input NAND gate. [6M]
- 4(a) Explain about clocking disciplines. [6M]
(b) Explain channel gate arrays. [6M]
(OR)
(c) What are problems presented by power distribution? How they are solved. [6M]
(d) Explain a pseudo static RAM cell with read and write operations. [6M]
- 5(a) What are various floor planning methods? Explain them clearly? [6M]
(b) Explain about system design methodologies. [6M]
(OR)
(c) Explain system-on-chip concept using platform based design. [6M]
(d) Write notes on Architecture testing of VLSI design. [6M]

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M.Tech. I Semester Regular/Supplementary Examinations

23 DEC 2013

MICROCONTROLLERS FOR EMBEDDED SYSTEM DESIGN

(Systems and Signal Processing (ECE))

Time : 3 hours

Max. Marks: 60

Answer all the questions

All Questions carry equal marks

- 1(a) Describe the classification of embedded systems. [6M]
(b) Draw the block diagram of embedded system hardware and explain each. [6M]
- (OR)**
- (c) Explain the design process in embedded system in detail. [6M]
(d) What are steps to be followed in selection of processor for an embedded system. [6M]
- 2(a) Describe the architecture of 8051 Microcontroller with neat block diagram. [8M]
(b) Explain the various modes of operation of counters in 8051 microcontroller. [4M]
- (OR)**
- (c) Describe the operation of memory and I/O addressing of 8051 microcontroller [8M]
(d) Write a short notes on PIC controller [4M]
- 3(a) Explain the architecture of RISC processor with neat sketch. [6M]
(b) Explain the way of programming of PSOC [6M]
- (OR)**
- (c) Explain the modes of operation of ARM processor and list out few instructions with examples. [8M]
(d) List out the applications of timer blocks in RISC processors [4M]
- 4(a) Explain about hardware and software interrupt sources and how to handle these interrupts. [10M]
(b) Describe the classification of interrupts. [2M]
- (OR)**
- (c) Describe the turns
(i) context (ii) latency (iii) Dead line [4M]
(d) Describe the operation of device driver using interrupt service Routine. [8M]
- 5(a) Explain SDMA? [6M]
(b) Explain the serial communication protocol [6M]
- (OR)**
- (c) Explain about external bus interface [6M]
(d) Explain Ethernet protocol [6M]